

The Pasting Lemma (§1.62)

by

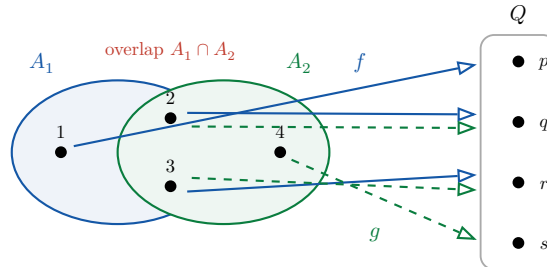
§1.62 — the intersection $\rightarrow$ union square of two subobjects is a pushout, with a concrete Set example and Freyd’s relational proof that R is a map

In a pre-logos, for subobjects $A_1, A_2 \hookrightarrow A$ with intersection $A_1 \cap A_2$ and union $A_1 \cup A_2$, this square of monics is a **pushout**:

$$\begin{array}{ccc}
 A_1 \cap A_2 & \xrightarrow{\bar{y}} & A_1 \\
 \bar{x} \downarrow & & \downarrow x \\
 A_2 & \xrightarrow{y} & A_1 \cup A_2
 \end{array}$$

Pushout means: a map $f : A_1 \rightarrow Q$ **from** A_1 and a map $g : A_2 \rightarrow Q$ **from** A_2 that **agree on the overlap** — i.e. their two restrictions to $A_1 \cap A_2$ coincide, $\bar{y}f = \bar{x}g$ (the top square commutes) — glue to a **unique** $R : A_1 \cup A_2 \rightarrow Q$ with $xR = f$ and $yR = g$.

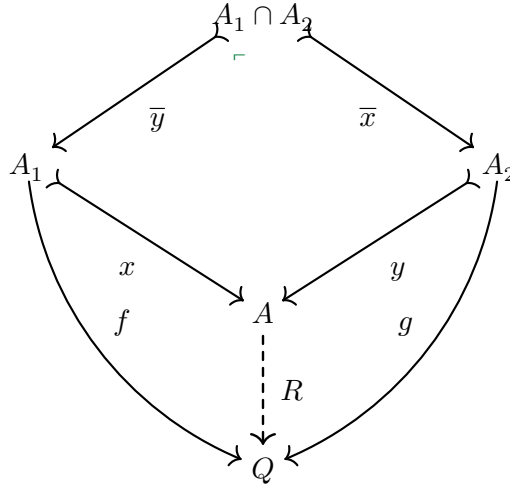
Concretely in SET: $A_1 = \{1, 2, 3\}$, $A_2 = \{2, 3, 4\}$, overlap $A_1 \cap A_2 = \{2, 3\}$, union $\{1, 2, 3, 4\}$, and $Q = \{p, q, r, s\}$. Take $f : 1 \mapsto p, 2 \mapsto q, 3 \mapsto r$ and $g : 2 \mapsto q, 3 \mapsto r, 4 \mapsto s$. The agreement is the commuting condition: on the overlap f and g **both** send $2 \mapsto q$ and $3 \mapsto r$ (the converging arrows below), so the four arrows form one well-defined $R : \{1, 2, 3, 4\} \rightarrow Q$. Had they disagreed at 2 or 3, no single R could exist — that is the whole force of “agree on the overlap”.



The pasted map $R = f \cup g$ is just **all six arrows together**; it is a well-defined function exactly because the blue (f) and green (g) arrows **coincide** on the overlap $\{2, 3\}$.

§1. The figure the book actually draws

Freyd’s proof is purely **relational** (it never names elements). The picture below is the one in the book on p.101: the pullback square on top, the two subobjects x, y landing in the union A , a test object Q with the agreeing pair f, g , and the mediator $R : A \rightarrow Q$ to be constructed. Bars name the **pullback legs** — $\bar{y}$ is the copy of y pulled back over A_1 , $\bar{x}$ the copy of x over A_2 — so $\bar{y} : A_1 \cap A_2 \rightarrow A_1$ and $\bar{x} : A_1 \cap A_2 \rightarrow A_2$.



Everything is forced by ten relational facts, exactly the list the book writes down (composition in diagram order, $R^\circ = \text{converse}$). The first eight are **generic regular-category** facts; only the cover identity needs the pre-logos.

- | | | | |
|--|---------------------------|--|-------------------------------------|
| (1) $xx^\circ = 1, yy^\circ = 1$ | — x, y monic | (2) $xy^\circ = \bar{y}^\circ \bar{x}$ | — pullback tabulates overlap |
| (3) $1 \subseteq ff^\circ, 1 \subseteq gg^\circ$ | — f, g entire | (4) $f^\circ f \subseteq 1, g^\circ g \subseteq 1$ | — f, g simple |
| (5) $\bar{x}g = \bar{y}f$ | — agree on overlap | (6) $(\bar{y}f)^\circ(\bar{y}f) \subseteq 1$ | — $\bar{y}f$ simple (★) |

Cover (the one pre-logos input): $x^\circ x \cup y^\circ y = 1$. A monic z contains $\text{Im}(x), \text{Im}(y)$ iff $x^\circ x \subseteq z^\circ z$ and $y^\circ y \subseteq z^\circ z$; in a pre-logos the union $A_1 \cup A_2$ is least, so its coreflexive is 1.

Why is (6) true? The book states it without a word. Because $\bar{y}f$ is a **composite of maps**: $\bar{y} : A_1 \cap A_2 \hookrightarrow A_1$ is monic (hence a map) and $f : A_1 \rightarrow Q$ is a map, and maps are closed under composition [§1.561]. Every map is **simple** [§1.564], i.e. satisfies $h^\circ h \subseteq 1$ — so $(\bar{y}f)^\circ(\bar{y}f) \subseteq 1$. That is the **entire content** of the unexplained line; the same remark gives $f^\circ f, g^\circ g \subseteq 1$ in (4).

Why the cover $x^\circ x \cup y^\circ y = 1$? For a subobject $x : A_1 \hookrightarrow A$, the relation $x^\circ x : A \rightarrow A$ is the **coreflexive** (sub-identity) cutting out $\text{Im}(x) = A_1$: in SET, $x^\circ x = \{(a, a) : a \in A_1\}$, the diagonal on $A_1 \subseteq A$. Likewise $y^\circ y$ is the diagonal on A_2 , and their union is the diagonal on $A_1 \cup A_2$. A monic z contains both images iff $x^\circ x \subseteq z^\circ z$ **and** $y^\circ y \subseteq z^\circ z$, i.e. iff $x^\circ x \cup y^\circ y \subseteq z^\circ z$ — so $x^\circ x \cup y^\circ y$ is the coreflexive of the **least** subobject containing A_1 and A_2 , their union $A_1 \cup A_2$. The proof's object A is that union (the pushout corner, where x, y land — as in the figure above), so its coreflexive is the whole identity 1_A . It is the **pre-logos** that makes the union of subobjects exist and be computed by this relational $\cup$.

Why $xy^\circ = \bar{y}^\circ \bar{x}$ (fact (2))? It is the **tabulation** of the composite — **by construction**, not a universal-property argument. Relational composition $x; y^\circ$ is **defined** by pulling back the cospan $A_1 \xrightarrow{x} A \xleftarrow{y} A_2$; that pullback **is** the intersection $A_1 \cap A_2$, and its two projections **are** the legs $\bar{y}, \bar{x}$. Reading the relation off that jointly-monic span gives the equality $xy^\circ = \bar{y}^\circ \bar{x}$ (likewise $yx^\circ = \bar{x}^\circ \bar{y}$). Lean's `inter_lemma` records only the containment $xy^\circ \subseteq \bar{y}^\circ \bar{x}$ —

all the monotone chain needs — because it forms the intersection over the object A_1, A_2 are subobjects of rather than over their union, so it compares two pullbacks by a map and lands on $\subseteq$; over the union (the figure's A) it is the equality above.

§2. The proof that R is a map (Freyd, p.101)

Do not build a function and verify “well-defined + agrees”. Instead write down a single **relation**

$$R = x^\circ f \cup y^\circ g : A \rightarrow Q$$

and prove it is a **map**. By §1.564 a relation is a map iff it is **entire** ($1 \subseteq RR^\circ$, total) and **simple** ($R^\circ R \subseteq 1$, single-valued). Both fall out of the facts above — totality from the cover, single-valuedness from the agreement — with the only pre-logos input being the cover and §1.616 (union distributes over composition, used to multiply out the unions).

Entire. Pump each cover-branch up with the totality of f, g (fact (3)), then factor:

$$\begin{aligned} 1 &\subseteq x^\circ 1x \cup y^\circ 1y \subseteq x^\circ f f^\circ x \cup y^\circ g g^\circ y \\ &\subseteq (x^\circ f \cup y^\circ g)(f^\circ x \cup g^\circ y) \subseteq RR^\circ. \end{aligned}$$

Simple. Multiply out (§1.616), kill the diagonal monic loops by (1), reroute the cross terms by the tabulation (2), and fold them together by the agreement (5):

$$\begin{aligned} R^\circ R &\subseteq (f^\circ x \cup g^\circ y)(x^\circ f \cup y^\circ g) \\ &\subseteq f^\circ x x^\circ f \cup f^\circ x y^\circ g \cup g^\circ y x^\circ f \cup g^\circ y y^\circ g \\ &\subseteq f^\circ f \cup f^\circ \bar{y}^\circ \bar{x} g \cup g^\circ \bar{x}^\circ \bar{y} f \cup g^\circ g \\ &\subseteq 1 \cup (\bar{y}f)^\circ \bar{x} g \cup (\bar{x}g)^\circ \bar{y} f \cup 1 \\ &\subseteq 1 \cup (\bar{y}f)^\circ \bar{y} f \cup (\bar{y}f)^\circ \bar{y} f \subseteq 1. \end{aligned}$$

Since $1 \subseteq RR^\circ$ and $R^\circ R \subseteq 1$, **R is a map** [§1.564]. It is the mediator: by (1), (2), (5),

$$xR = x(x^\circ f \cup y^\circ g) = x x^\circ f \cup x y^\circ g = f \cup \bar{y}^\circ \bar{x} g = f \cup \bar{y}^\circ \bar{y} f = (1 \cup \bar{y}^\circ \bar{y})f = f,$$

and symmetrically $yR = g$. Uniqueness of R is the cover once more: x, y are jointly epic.

★ Punchline

The mediating map is forced by the relational calculus: $R = x^\circ f \cup y^\circ g$ is **entire** because the cover splits 1_A and the totality of f, g pushes each branch into RR° ; **simple** because $R^\circ R$ multiplies out into four terms that each reduce to $\subseteq 1_Q$ (the diagonals by the monic $xx^\circ = 1$ and the map-cap $f^\circ f \subseteq 1$; the cross terms by the tabulation $xy^\circ = \bar{y}^\circ \bar{x}$, the agreement (5), and the map-cap of $\bar{y}f$); and **unique** because x, y are jointly epic. The one step that needs the pre-logos is $\cup$ distributing over composition (§1.616) — which is why the lemma fails additively (1.622: it breaks for groups).

§3. §1.622 — when pasting can fail

The §1.62 proof has one non-formal input: $\cup$ distributes over composition (§1.616). Where that fails, so can the lemma. §1.622 records two faces of this.

Abelian categories — still a pushout, but for a different reason. There $A_1 \cup A_2 = A_1 + A_2$, and the intersection $\rightarrow$ union square is **bicartesian** (pullback = pushout) straight from the additive structure (the second isomorphism theorem). Freyd’s relational proof does **not** apply: unions of relations do not distribute over composition in any non-degenerate additive regular category.

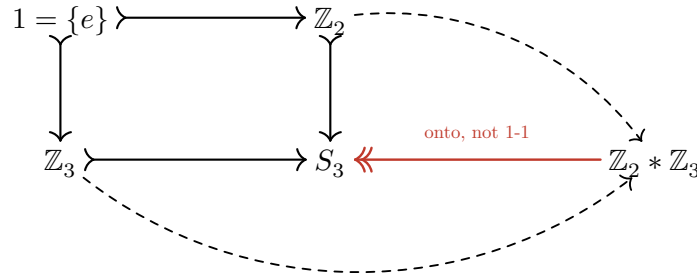
Groups — not a pushout. S_3 is the group of all **six** rearrangements of three corners 0, 1, 2 (the symmetries of a triangle): three rotations 012 (do nothing), 120, 201, and three swaps 102, 021, 210 (each list records where corners 0, 1, 2 go). Inside it sit two subgroups:

- $A_1 = \mathbb{Z}_2 = \{012, s\}$, the identity and the swap $s = 102$ — order 2, since s twice is the identity;
- $A_2 = \mathbb{Z}_3 = \{012, r, r^2\}$, the identity and the rotation $r = 120$ — order 3.

The meet $A_1 \cap A_2 = 1$. An intersection of subgroups is a subgroup of each, so by Lagrange its size divides both 2 and 3 — the only common divisor is 1, leaving just the identity 012 (the one list in both). Coprime orders force a trivial overlap.

The union $A_1 \cup A_2 = S_3$ — and this is the crux. The lattice “union” is **not** set-union: $\{012, 102, 120, 201\}$ is not even a subgroup, since composing s with r gives 021, outside the list. The **join** is the **smallest subgroup containing both** — the subgroup **generated** by s and r (close that set under composition). A swap and a rotation generate everything, so $\langle s, r \rangle = S_3$ (its order is divisible by 2 and 3, hence by $6 = |S_3|$).

This closing-under-composition is exactly where groups depart from SET: there the join of two **subsets** simply **is** their set-union, with nothing to close — so pasting always works. §1.62 would make the square below a pushout — it is not.



The true pushout of $\mathbb{Z}_2 \leftarrow 1 \rightarrow \mathbb{Z}_3$ in GRP is the **free product** $\mathbb{Z}_2 * \mathbb{Z}_3$ — the **infinite** modular group $\text{PSL}_2(\mathbb{Z})$. The comparison $\mathbb{Z}_2 * \mathbb{Z}_3 \twoheadrightarrow S_3$ is onto but far from injective: S_3 is the proper quotient that imposes the dihedral relation $srs = r^2$ (a swap conjugates the rotation to its inverse), which the free product never sees. A finite group is not the gluing; the genuine gluing is infinite.

Pasting literally fails here. Put $Q = \mathbb{Z}_2 * \mathbb{Z}_3$ and let $f : \mathbb{Z}_2 \rightarrow Q$ and $g : \mathbb{Z}_3 \rightarrow Q$ be the two inclusions. They **agree on the overlap** 1 (both fix e) — yet **no** homomorphism $R : S_3 \rightarrow Q$ extends them: its image would be a **finite** subgroup of Q containing both factors, but those two already generate the **infinite** Q . Two maps agreeing on the overlap with nothing to glue them onto: exactly the failure of “paste”. In SET (or any pre-logos) this cannot happen — §1.616 guarantees the gluing exists and is the union.

§4. §1.623 — positive pre-logos: sums are tagged unions

A **positive pre-logos** is a pre-logos in which every pair of objects A, B embeds **disjointly** into some common object. The book’s whole definition is one square:

$$\begin{array}{ccc} 0 & \xrightarrow{\quad} & B \\ \downarrow & \lrcorner & \downarrow \\ A & \xrightarrow{\quad} & C \end{array}$$

Reading the square. The corner mark at 0 makes it a **pullback**, and the pullback of two monos $A \hookrightarrow C \leftarrow B$ is by definition their intersection in $\text{Sub}(C)$. So the diagram says: $A \cap B = 0$ inside C — the two embedded copies are disjoint.

The quantifier is $\exists$, not $\forall$. It does **not** say every pair of subobjects of every C meets in 0 — that is false in any interesting category (take $A = B = C$: the pullback is A itself). It says: for every pair of **objects** there **exists some** host C into which they embed disjointly. Choosing the least such host, $A \cup B = C$, gives the coproduct — hence a positive pre-logos has coproducts.

This is Haskell’s `Either a b = Left a | Right b`. In **SET** the host is the tagged union $C = (\{0\} \times A) \cup (\{1\} \times B)$: the tag is what forces the pullback to be empty even when $A = B$, since $(0, a)$ never equals $(1, a)$. And it is exactly why “has coproducts” is **weaker** than “positive”: a lattice has no tags, so its join collapses the overlap $A \wedge B$ instead of separating it — §1.626 below works this out on a concrete lattice.

§1.624 is pattern matching. Given $f : X \rightarrow B_1 + B_2$, set $X_i = f^\#(B_i)$ — “the inputs that hit **Left**” and “the inputs that hit **Right**”:

$$\begin{array}{ccccc} X_1 & \xrightarrow{\quad} & X & \xleftarrow{\quad} & X_2 \\ \downarrow f_1 & & \downarrow f & & \downarrow f_2 \\ B_1 & \xrightarrow{\quad} & B_1 + B_2 & \xleftarrow{\quad} & B_2 \end{array}$$

Then $X = X_1 + X_2$ and $f = f_1 + f_2$: every map into a sum factors through a case analysis of its source. The case split is **exhaustive** because $f^\#$ preserves unions (the pre-logos axiom): $X_1 \cup X_2 = f^\#(B_1 \cup B_2) = f^\#(B_1 + B_2) = X$. It is **exclusive** — no input matches both branches — by the §1.61 toolkit: $f^\#$ preserves $\cap$ for free (pullbacks commute), and $f^\#(0) = 0$ is the nullary-union axiom, so $X_1 \cap X_2 = f^\#(B_1 \cap B_2) = f^\#(0) = 0$. Finally, disjoint ($\cap = 0$) + covering ($\cup = X$) subobjects form a coproduct — that is the corollary of the pasting lemma quoted in §1.622 above.

★ Punchline

Positivity is disjunction **with tags**: landing in $A + B$ tells you which summand you are in. A pre-logos interprets the positive fragment of first-order logic ($\top, \wedge, \perp, \vee, \exists$ — no $\neg, \Rightarrow, \forall$) on its subobject lattices; a **positive** pre-logos realizes that fragment’s $\vee$ at the object level, as **Either** — disjoint, stable, pattern-matchable sums, the way **SET** has them and a lattice of truth values does not.

§5. §1.625 — a regular representation is a pre-logos one iff it preserves disjoint unions

Suppose A and B are positive pre-logoi and that $T : A \rightarrow B$ is a representation of regular categories. Then T is a representation of pre-logoi iff it preserves disjoint unions.

A **representation of regular categories** preserves finite limits and images (covers). A **representation of pre-logoi** must preserve one more thing: the pre-logos structure — the finite **unions** of subobjects $A_1 \cup A_2$ (and the empty union 0). (One trap: “ $\cup$ is stable under inverse image” is an **axiom of the category A** ; it is **not** a condition on T . What T has to preserve is the **operation $\cup$ itself**.) So the entire gap between the two notions is one question: **does T preserve $\cup$?**

Freyd’s one-line “because”: a union is the image of a coproduct.

$$A_1 \cup A_2 = \text{Im}([i_1, i_2] : A_1 + A_2 \rightarrow A).$$

The copairing $[i_1, i_2]$ lands both summands in A ; its image — the least subobject containing both — is their union. It factors as cover-then-mono:

$$\begin{array}{ccc} A_1 + A_2 & \xrightarrow{[i_1, i_2]} & A \\ & \searrow \text{cover} & \nearrow \\ & A_1 \cup A_2 & \end{array}$$

In **SET**, with $A_1 = \{1, 2, 3\}$ and $A_2 = \{3, 4, 5\}$ inside $A = \{1, 2, 3, 4, 5\}$: the coproduct $A_1 + A_2$ is the **6-element tagged union**, the copairing **fuses** the two copies of 3, and the **image** of that fusing is exactly $A_1 \cup A_2 = \{1, 2, 3, 4, 5\}$ (5 elements). Coproduct, then image, gives union.

The union is embedding-dependent; the coproduct is not. The ambient A is only a **container** — it need not be disjoint, nor even equal $A_1 \cup A_2$ (A may be far bigger; the image still carves the union out). What varies is the **overlap** $A_1 \cap A_2$ inside A : the copairing is **monic** $\iff A_1 \cap A_2 = 0$ (in **SET** the only collisions are shared elements). So a union is **the coproduct with that overlap fused in** — a disjoint embedding fuses nothing, and there $\cup = +$ (§1.621). The **same** two sets get **different** unions under different embeddings, while the coproduct never moves:

$A_1 = \{1, 2, 3\}$, and B in A	$A_1 \cup B$	vs. coproduct
$B = \{a, b\}$ (disjoint)	$\{1, 2, 3, a, b\}$ (5)	= coproduct
$B = \{3, 4\}$ (share 3)	$\{1, 2, 3, 4\}$ (4)	< coproduct
$B = \{2, 3\}$ ($B \subseteq A_1$)	$\{1, 2, 3\}$ (3)	< coproduct

The coproduct is always the 5-element $\{1, 2, 3\} + B$; the union slides between $\max(|A_1|, |B|)$ and $|A_1| + |B|$ as the overlap shrinks. $\text{Im}(A_1 + A_2 \rightarrow A)$ is the coproduct pushed through the chosen embedding.

Why this settles the “iff”. T already preserves images, so

$$T(A_1 \cup A_2) = T(\text{Im}(A_1 + A_2 \rightarrow A)) = \text{Im}(T(A_1 + A_2) \rightarrow TA).$$

- If T preserves the **coproduct**, i.e. $T(A_1 + A_2) = TA_1 + TA_2$, the right side is $\text{Im}(TA_1 + TA_2 \rightarrow TA) = TA_1 \cup TA_2$: **preserves disjoint unions $\Rightarrow$ preserves unions**.
- Conversely T preserves pullbacks and 0, so $A_1 \cap A_2 = 0$ forces $TA_1 \cap TA_2 = 0$; with union-preservation, $T(A_1 + A_2) = TA_1 \cup TA_2$ is then a **disjoint union** $= TA_1 + TA_2$: **preserves unions $\Rightarrow$ preserves disjoint unions**.

Concrete example. The evaluation functor $\text{ev}_a : \mathcal{S}^{\mathcal{A}} \rightarrow \mathcal{S}$ ($\mathcal{S} = \text{SET}$, so $\mathcal{S}^{\mathcal{A}}$ is the functor category $\mathcal{A} \rightarrow \text{Set}$), $F \mapsto F(a)$, is a representation of regular categories — limits, covers and images in $\mathcal{S}^{\mathcal{A}}$ are all computed pointwise. It preserves disjoint unions, $\text{ev}_a(F + G) = (F + G)(a) = F(a) + G(a)$, so by §1.625 it is a representation of pre-logoi; and indeed $(F \cup G)(a) = F(a) \cup G(a)$ — the coproduct-then-image at a is exactly the coproduct-then-image in $\mathcal{S}$.

★ Punchline

You never check union-preservation directly. A regular representation already carries images, and **every union is the image of a coproduct**, so union-preservation collapses to the one cheaper condition “ T preserves disjoint unions”. The condition has teeth: §1.626 (below) notes that the canonical embedding of a pre-logos into a positive one [2.217] is a regular representation that is **not** a pre-logos one — precisely because it fails to preserve coproducts.

§6. §1.626 — coproducts without positivity

Coproducts can exist without positivity. Any distributive lattice, viewed as a category is a pre-logos with coproducts. It is positive iff it is degenerate.

The point is that “coproduct” is a **universal property** — a map $A + B \rightarrow Q$ is exactly a pair $A \rightarrow Q, B \rightarrow Q$ — which **never mentions disjointness**. So $A + B$ can exist while the two injections are **not** disjoint. Positivity is that missing extra: $A \cap B = 0$ (injections monic and jointly covering). A category can have every coproduct and still fail it.

Witness. Any distributive lattice, viewed as a thin category (one arrow $a \rightarrow b$ iff $a \subseteq b$), has products $= \cap$ and coproducts $= \vee$: the least upper bound $a \vee b$ genuinely satisfies the coproduct universal property. So coproducts exist — but never disjointly:

$$\{1, 2\} + \{1, 2\} = \{1, 2\} \vee \{1, 2\} = \{1, 2\}, \quad \text{hence} \quad \{1, 2\} \cap \{1, 2\} = \{1, 2\} \neq 0.$$

Both copies collapse onto one; contrast SET, where $1 + 1 = 2$ keeps them tagged apart.

Positive iff degenerate. Taking $b = a$, positivity would force $a \cap a = a = 0$ for **every** object a — collapsing the whole lattice to a single point $0 = 1$.

★ Punchline

This is exactly why the Lean class `DisjointBinaryCoproduct` (Fredy/S1_62.lean, §1.62) carries axioms **beyond** `HasBinaryCoproducts`: the lattice satisfies `HasBinaryCoproducts` — its join is a coproduct — yet violates `inl n inr ≤ 1` (`inl_inter_inr`). Positivity is the

“not-a-mere-lattice” content: disjoint, tagged sums, the way SET has them and a lattice of truth values does not.